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A
Project Report
on
SHOPLIFTING DETECTION MODEL
SHOPGUARD
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MAY , 2025

DECLARATION

Declaration

We hereby declare that this submission is our work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material that has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

The work presented in this report is original and has been carried out for academic purposes. Any references to existing research, frameworks, or methodologies have been properly cited.

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CERTIFICATE

This is to certify that the Project Report entitled “**Shop Shoplifting Detection: A Real-Time Surveillance System**” which is submitted by Student name in partial fulfillment of the requirement for the award of degree B. Tech. in the Department of Computer Science & Engineering of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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We also do not like to miss the opportunity to acknowledge the contribution of all faculty members, especially faculty/industry person/any person, of the department for their kind assistance and cooperation during the development of our project. Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

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ABSTRACT

Shoplifting poses a major challenge to the retail industry, causing significant financial losses. This study presents a real-time shoplifting detection framework using a hybrid **Convolutional Long Short-Term Memory (ConvLSTM)** network. The model extracts spatial features from video frames via convolutional layers and captures temporal dependencies using LSTM layers, enabling accurate detection of complex shoplifting behaviors.

The **Adam optimizer** is employed to enhance training efficiency by dynamically adjusting learning rates. Experiments on a shoplifting video dataset show that our model achieves **83% accuracy**, outperforming baseline methods. We also discuss challenges like dataset imbalance and environmental sensitivity, along with potential improvements such as data augmentation and attention mechanisms. This research contributes to developing automated, scalable shoplifting detection systems for retail environment.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
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CNN	Convolutional Neural Network
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DL	Deep Learning
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FPS	Frames Per Second
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GIS	Geographic Information System
------------	-------------------------------

LSTM	Long Short-Term Memory
-------------	------------------------

ML	Machine Learning
-----------	------------------

NLP	Natural Language Processing
------------	-----------------------------

RNN	Recurrent Neural Network
------------	--------------------------

ROI	Region of Interest
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CHAPTER 1 INTRODUCTION

1.1 INTRODUCTION

Shoplifting, which involves taking goods from a retail store without paying, has long been a significant challenge in the global retail sector. Since the rise of trade in medieval markets, store owners have actively sought ways to prevent and identify minor theft. With the rise of modern consumer capitalism and the growth of large retail establishments, shoplifting has transformed from sporadic minor theft to a coordinated and systematic danger.

During the 20th century, the introduction of self-service shopping options—like supermarkets, department stores, and big-box retailers—allowed customers more freedom but also decreased direct supervision from staff. This change, while beneficial

for economic efficiency, also made retail stores more susceptible to theft. Research from the National Association for Shoplifting Prevention (NASP) indicates that a shoplifter is caught only once for every 48 incidents on average, highlighting the extent of unreported crime.

1.1.1 Economic Impact

Retail theft results in billions of dollars in losses each year on a global scale. The Global Retail Theft Barometer (2022) reports that retailers worldwide experience total losses exceeding \$100 billion annually because of theft, with shoplifting responsible for more than 35% of that amount. This results in higher prices for consumers, reduced profit margins for companies, and increased spending on security measures.

In India, retail theft is largely underreported but is believed to lead to annual losses between ₹5,000 and ₹7,000 crore, based on industry surveys. These losses are especially detrimental to small and medium enterprises, where even small theft incidents can greatly impact their viability.

In addition to direct financial losses, there are also indirect effects:

- Erosion of customer trust resulting from false accusations or ineffective monitoring.
- Higher operational expenses due to hiring security staff.
- Diminished employee morale caused by a culture of pressure and blame.

1.1.2 Traditional Detection Methods: Their Limitations

Traditionally, retailers have depended on a mix of physical deterrents (such as tags and alarms) and human oversight (like security personnel or CCTV surveillance).

Nevertheless, human error, fatigue, and limited attention spans can diminish the effectiveness of manual monitoring, particularly in stores that have numerous camera feeds and high levels of customer traffic.

Reviewing video footage manually is labor-intensive, subjective, and variable. Most instances of shoplifting go unnoticed in real time, making it extremely difficult to recover stolen items or swiftly apprehend the thief. This underscores the pressing need for automated, real-time surveillance systems that can help security teams proactively detect and highlight suspicious activities.

4.1.2 Rise of AI and Deep Learning in Retail Security

The rise of Artificial Intelligence (AI) and Deep Learning (DL) has transformed the functionality of contemporary surveillance systems. Through the use of models like

Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, machines are now capable of identifying patterns, spotting anomalies, and making informed decisions based on visual information.

In particular, ConvLSTM (Convolutional Long Short-Term Memory) models are particularly effective for spatiotemporal data such as video surveillance. They can evaluate both spatial characteristics (what is present in the frame) and temporal dynamics (how things evolve over time), making them optimal for identifying suspicious behavior patterns rather than just single frames.

1.1.3 Research Relevance

This initiative is centered on creating a real-time detection system for shoplifting that

utilizes a ConvLSTM-based Deep Learning approach. By integrating both spatial and temporal analysis of video footage, the system seeks to:

Lessen dependence on human oversight.

Allow for immediate notifications regarding questionable behavior.

Reduce instances of false positives by identifying behavioral patterns.

Tackle real-world challenges such as class imbalance and variations in lighting within surveillance videos.

1.2 PROJECT DESCRIPTION

The main objective of this project is to create a shoplifting detection system based on Machine Learning, which utilizes Convolutional Long Short-Term Memory

(ConvLSTM) models to identify incidents of shoplifting in real-time using video surveillance. This system is intended to serve as a smart, automated solution that enhances traditional security methods with the intelligence and adaptability of advanced artificial intelligence algorithms.

The proposed system makes use of live video footage collected from CCTV or IP cameras placed in retail settings. By continuously analyzing the video streams, the model detects abnormal or suspicious behaviors typically linked to theft, such as sudden concealment of items, erratic hand movements, or frequent looks toward security exits.

1.2.1 Core Idea and Approach

The proposed solution seeks to automatically identify shoplifting behavior by training a deep learning model on a labeled dataset that includes both typical and shoplifting

scenarios, in contrast to traditional systems that depend on human guards to monitor security feeds. This model analyzes video data to comprehend not only individual frames (spatial characteristics) but also the progression of actions over time (temporal dependencies).

The ConvLSTM architecture is exceptionally well-suited for this purpose because:

Convolutional layers identify spatial characteristics such as body posture, hand actions, and the position of items.

LSTM layers capture the dynamics over time, discerning whether a sudden movement relates to a suspicious pattern or represents a normal customer behavior.

By integrating these two learning approaches, the model achieves a comprehensive understanding of customer behavior throughout time.

1.2.1.1 Automated Detection

The primary objective is to either replace or enhance traditional surveillance with an AI-driven system capable of autonomously detecting shoplifting behavior. The system is designed to identify a range of questionable actions, including:

- Frequent checks of the environment.
- Hiding items beneath clothing or inside bags.
- Abrupt, atypical body movements.

By utilizing supervised learning on labeled video sequences, the model is trained to differentiate between normal behaviors and potential shoplifting incidents with a high degree of precision.

1.2.1.2 Real-Time Analysis

Real-time processing is essential for the practical functionality of the system. The system is required to:

- Handle ongoing video feeds from surveillance cameras.

- Identify suspicious behavior with very low latency.
- Activate alerts to notify security personnel or staff as soon as a suspicious behavior pattern is detected.

This enables staff to take action during or right after an attempted theft, enhancing the likelihood of recovery and discouragement.

1.2.1.3 Enhanced Security

This system is designed to:

- Minimize human mistakes that arise from fatigue or lapses in attention during extended monitoring.
- Enhance scalability—allowing a single system to oversee numerous video streams at the same time.
- Augment incident documentation by saving prediction records, timestamps, and video footage of potential incidents.

By transitioning from a reactive security method to active detection, the system changes the way retailers handle loss prevention.

1.2.2 Functional Workflow

- Video Input

Live video is obtained from the current in-store surveillance systems.

- Preprocessing & Frame Extraction

The video is divided into consistent frame sequences. The frames are resized and normalized for model input.

- ConvLSTM Inference

The model conducts spatiotemporal analysis on the sequences to identify patterns.

- Suspicion Score Prediction

Each segment receives a “risk score” or is categorized as normal/shoplifting.

- Alert Generation

If a shoplifting prediction is made, the system sends a real-time alert and can optionally capture a snapshot for security assessment.

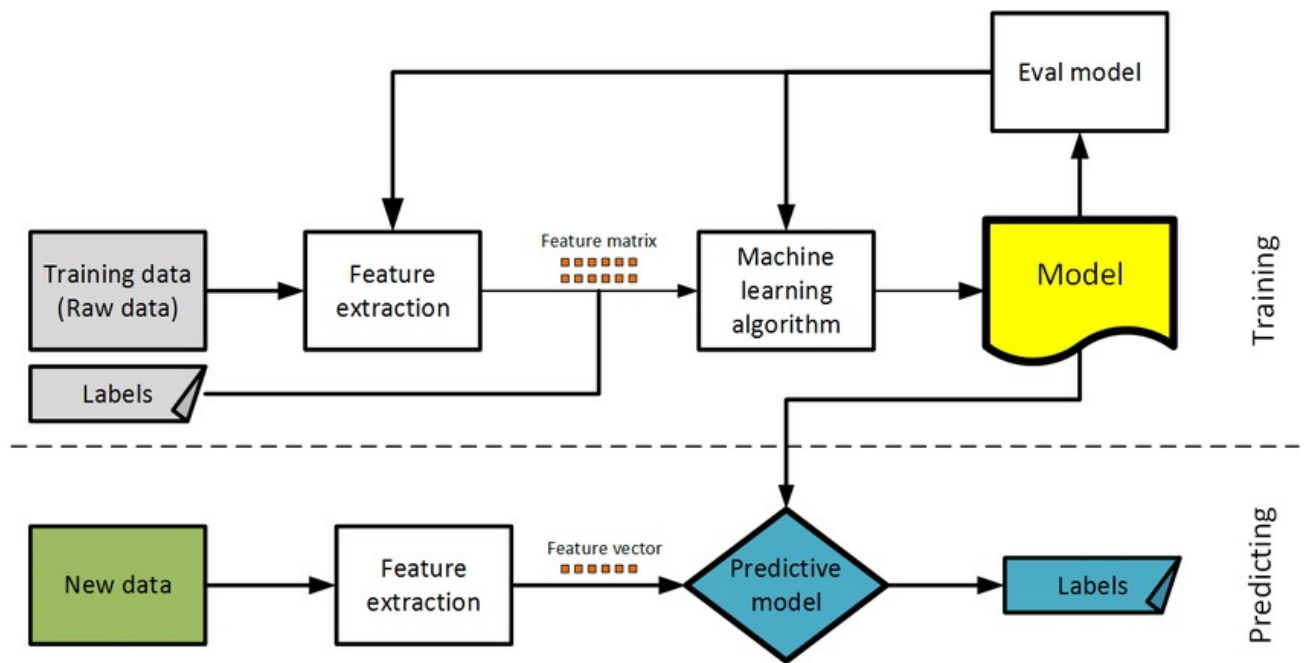


Fig 1 Flow chart of the model

CHAPTER 2 LITERATURE REVIEW

As retail spaces evolve and modernize, the challenges associated with shoplifting are becoming increasingly intricate, discreet, and harder to handle with conventional monitoring methods. The use of machine learning—especially deep learning—has gained substantial traction in the domain of video-based security. This chapter examines pertinent research and technologies that contribute to the design and advancement of the existing shoplifting detection system.

2.1 Deep Learning Techniques Used in Retail

Deep learning techniques have found increasing applications in retail security, particularly in automating monitoring and enhancing the detection of shoplifting.

These approaches are generally categorized into three main types:

4. Convolutional Neural Networks (CNNs)

CNNs serve as the foundation for computer vision systems and are widely employed to classify objects, recognize individuals, and track item movements in surveillance videos. However, CNNs focus on analyzing individual frames, which limits their ability to model behaviors that develop over time.

Typically, CNNs are utilized alongside object detection approaches such as YOLO (You Only Look Once), SSD (Single Shot Multibox Detector), or Faster R-CNN to monitor objects and detect suspicious hand movements.

2. Recurrent Neural Networks (RNNs) and LSTM

RNNs, particularly Long Short-Term Memory (LSTM) networks, excel at modeling

sequential data. Within the realm of surveillance, they aid in identifying irregular activities over time, such as repeated visits to the same aisle or extended presence near a shelf.

Nonetheless, LSTMs do not have the capacity to interpret visual characteristics in each frame, making them less effective when used in isolation.

3. ConvLSTM (Convolutional Long Short-Term Memory)

ConvLSTM merges the advantages of both CNNs and LSTMs by facilitating spatial modeling (through convolution) and temporal modeling (through memory gates) in video data. This model is especially ideal for surveillance, as it captures both what occurs and how it evolves over time.

ConvLSTM models analyze sequences of frames and track the changes in motion,

posture, or object locations, enabling the detection of intricate behaviors like concealment or theft.

4.1 Research Gaps and Industry Challenges

Despite advancements in automated surveillance, multiple research and practical obstacles remain:

- **Equipment and Performance Constraints**
 1. Real-time analysis leveraging deep learning necessitates significant computational resources (GPUs, edge AI devices), which may not be attainable for small or medium-sized retailers.
 2. Restrictions in frame rates and video quality can influence the speed and accuracy of

predictions.

- Imbalanced Datasets

1. Instances of shoplifting are significantly less frequent than typical retail operations.

This results in skewed datasets that can hinder models from effectively generalizing rare theft situations, leading to an increase in false positive rates.

- Concerns About Privacy and Ethics

1. AI-driven surveillance prompts privacy dilemmas, particularly in areas with stringent data protection regulations (e.g., GDPR in Europe).
2. Incorrect classifications could lead to false accusations, damaging customer confidence and the reputation of the store.

- Need for Contextual Awareness and Understandability

1. Current models frequently lack the ability to grasp context—such as distinguishing between a customer adjusting their clothing versus concealing an item.
2. Deep learning models are often regarded as black boxes, making it challenging for security personnel to interpret their decisions.

- Challenges in Scalability and Integration

1. Many solutions do not offer easy integration. They typically require specialized development to connect with existing CCTV systems, POS systems, or alert tools in real-time.
2. These shortcomings underscore the necessity for a solid, interpretable, lightweight model that can function in real-time with limited infrastructure while reducing ethical concerns.

4.1 Related Work Comparison Table

In order to grasp the trade-offs among various deep learning models employed for shoplifting or anomaly detection, the table below evaluates widely used architectures:

Research Topic	Year	Author	Contribution	Dataset	Evaluation Metrics
Deep learning-based anomaly detection	2023	Duong et al. [9]	Surveyed deep learning methods for video anomaly detection	Multiple public datasets	Accuracy, Precision, Recall
Spatiotemporal anomaly detection	2020	Nawaratne et al. [10]	Proposed ISTL framework for real-time surveillance	CUHK Avenue Dataset	Detection Rate, False Alarm Rate
Anomaly detection in surveillance	2022	Nithesh et al. [11]	Developed CNN-based deep learning model for anomaly detection	UCF-Crime Dataset	Accuracy, F1-score
AI-based anomaly detection	2024	Jain et al. [12]	Introduced CNN model for improved spatiotemporal anomaly recognition	ShanghaiTech Dataset	Precision, Recall, Accuracy
Shoplifting detection	2023	Muneer et al. [13]	Created a benchmark dataset and CNN-BiLSTM model	Custom Shoplifting Dataset	Accuracy, Sensitivity, Specificity
Contrastive attention for anomaly detection	2022	Chang et al. [4]	Designed a contrastive attention module for weakly supervised learning	UCSD Pedestrian Dataset	Precision, ROC-AUC Score
Attention-driven loss for anomaly detection	2020	Zhou et al. [5]	Introduced an attention-driven loss function for improved generalization	UMN Dataset	Detection Accuracy, F1-score
Influence-aware anomaly detection	2022	Zhang et al. [6]	Developed influence-aware attention model for surveillance	Subway Entrance Dataset	Precision, Recall, False Alarm Rate
Visual attention-based anomaly detection	2020	Nasaruddin et al. [7]	Applied spatiotemporal attention mechanism with 3D CNN	VIRAT Dataset	Accuracy, Computational Efficiency
CNN-BiLSTM for anomaly	2021	Ullah et al. [8]	Combined CNN and BiLSTM for	XD-Violence Dataset	F1-score, Sensitivity

detection			real-time video anomaly detection		
Autoencoder-based anomaly detection	2020	Ribeiro et al. [1]	Designed convolutional autoencoder for spatiotemporal anomaly detection	UCSD Ped2 Dataset	Precision, Detection Rate
Human action recognition in surveillance	2022	Dasari et al. [2]	Used ConvLSTM and LRCN for human activity recognition	Kinetics-600 Dataset	Accuracy, Recall, Confusion Matrix

Table 1 showing Previous Related work

CHAPTER 3 PROPOSED METHODOLOGY

3.1 Dataset

The project utilizes a dataset derived from video surveillance recordings captured in simulated retail settings. Each video clip depicts either typical shopping behavior or actions indicative of shoplifting. These videos have been manually annotated to facilitate supervised learning, wherein the model is trained to recognize and distinguish between behaviors by examining labeled instances.

- **Activity Type:** Normal behavior, shoplifting incidents.
- **Frame Sequences:** Extracted from videos at a uniform frame rate.
- **Annotations:** Labeled instances of shoplifting and normal activities
- The dataset includes **896 video sequences**, ensuring a balanced distribution of both classes.

Example data fields:

Frame ID	Activity Type	Timestamp	Motion Features	Suspicious Objects
001	Normal Activity	00:00:05	Low Movement	No
002	Shoplifting	00:00:12	Rapid Motion	Hidden Object

Table 2 showing Tie stamp

- **Characteristics of the Dataset:**

Total Number of Samples: 896 video sequences.

Class Distribution: The dataset is balanced, with roughly 741 labeled as “normal” and 155 categorized as “shoplifting/suspicious.”

Frame Dimensions: 128×128 pixels (uniformly resized for input into the model).

Duration per Sequence: Each video consists of 30 frames (equating to 2 seconds of footage at 15 FPS).

- **Data Annotation**

Accurate labeling of suspicious versus non-suspicious behaviors is essential for effective deep learning model training. The dataset was annotated with:

Motion vector analysis, which examines frame-to-frame variations in pixel intensity.

This thorough annotation process guarantees that the model acquires behavioral context rather than solely recognizing posture or objects.

3.2. Deep Learning Model Based on ConvLSTM

Why Opt for ConvLSTM?

ConvLSTM (Convolutional Long Short-Term Memory) is a combined deep learning framework that integrates:

Convolutional Neural Networks (CNNs) for extracting spatial features, along with LSTM (Long Short-Term Memory) units to capture temporal dynamics.

This structure is particularly well-suited for video data, as it processes sequences of frames while retaining contextual changes over time—an ideal approach for detecting suspicious action sequences that occur within a few seconds.

```
history = model.fit(x_train,
                    y_train,
                    epochs=30,
                    batch_size=4,
                    shuffle=True,
                    validation_split=0.2)
```

Run Cell | Run Above | Debug Cell

In[36]:

fig 2: Showing model training

```
1/135 ————— 14s 105ms/step - accuracy: 0.5000 - loss: 0.8719
W0000 00:00:1713897571.567354 3694 graph_launch.cc:671] Fallback to op-by-op mode because memset node breaks graph update
135/135 ————— 10s 72ms/step - accuracy: 0.8268 - loss: 0.4599 - val_accuracy: 0.7926 - val_loss: 0.5050
Epoch 3/50
135/135 ————— 10s 73ms/step - accuracy: 0.8310 - loss: 0.4527 - val_accuracy: 0.7926 - val_loss: 0.5052
Epoch 4/50
135/135 ————— 10s 72ms/step - accuracy: 0.8354 - loss: 0.4237 - val_accuracy: 0.8000 - val_loss: 0.4981
Epoch 5/50
135/135 ————— 10s 73ms/step - accuracy: 0.8288 - loss: 0.3823 - val_accuracy: 0.7704 - val_loss: 0.5244
Epoch 6/50
135/135 ————— 10s 73ms/step - accuracy: 0.8560 - loss: 0.3408 - val_accuracy: 0.7630 - val_loss: 0.5601
Epoch 7/50
135/135 ————— 10s 73ms/step - accuracy: 0.8573 - loss: 0.3160 - val_accuracy: 0.7481 - val_loss: 0.5946
Epoch 8/50
135/135 ————— 10s 72ms/step - accuracy: 0.8807 - loss: 0.2884 - val_accuracy: 0.7481 - val_loss: 0.6790
Epoch 9/50
135/135 ————— 10s 72ms/step - accuracy: 0.8748 - loss: 0.2713 - val_accuracy: 0.7630 - val_loss: 0.6196
Epoch 10/50
135/135 ————— 10s 72ms/step - accuracy: 0.8772 - loss: 0.2524 - val_accuracy: 0.7630 - val_loss: 0.5654
Epoch 11/50
135/135 ————— 10s 72ms/step - accuracy: 0.8940 - loss: 0.2691 - val_accuracy: 0.7481 - val_loss: 0.6442
Epoch 12/50
135/135 ————— 10s 72ms/step - accuracy: 0.8930 - loss: 0.2296 - val_accuracy: 0.8000 - val_loss: 0.5584
Epoch 13/50
135/135 ————— 10s 72ms/step - accuracy: 0.9123 - loss: 0.2005 - val_accuracy: 0.8074 - val_loss: 0.8302
Epoch 14/50
135/135 ————— 10s 73ms/step - accuracy: 0.9305 - loss: 0.1783 - val accuracy: 0.8000 - val loss: 0.8231
```

Fig 3 showing epoch training

```
convlstm_1 = ConvLSTM2D(filters=4, kernel_size=(3,3), activation='tanh', data_format='channels_last', recurrent_dropout=0.2)(input)
pool1 = MaxPooling3D(pool_size=(1,2,2), padding='same', data_format='channels_last')(convlstm_1)
timedistributed_1 = TimeDistributed(Dropout(0.2))(pool1)

convlstm_2 = ConvLSTM2D(filters=8, kernel_size=(3,3), activation='tanh', data_format='channels_last', recurrent_dropout=0.2)(timedistributed_1)
pool2 = MaxPooling3D(pool_size=(1,2,2), padding='same', data_format='channels_last')(convlstm_2)
timedistributed_2 = TimeDistributed(Dropout(0.2))(pool2)

convlstm_3 = ConvLSTM2D(filters=16, kernel_size=(3,3), activation='tanh', data_format='channels_last', recurrent_dropout=0.2)(timedistributed_2)
pool3 = MaxPooling3D(pool_size=(1,2,2), padding='same', data_format='channels_last')(convlstm_3)
timedistributed_3 = TimeDistributed(Dropout(0.2))(pool3)

convlstm_4 = ConvLSTM2D(filters=32, kernel_size=(3,3), activation='tanh', data_format='channels_last', recurrent_dropout=0.2)(timedistributed_3)
pool4 = MaxPooling3D(pool_size=(1,2,2), padding='same', data_format='channels_last')(convlstm_4)

flatten = Flatten()(pool4)

output = Dense(units=len(CLASSES_LIST), activation='softmax')(flatten)
```

Fig 4 showing Model used

```
import seaborn as sns

plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted labels')
plt.ylabel('True labels')
plt.title('Confusion Matrix')
plt.show()
```

Fig 5 showing heatmap plotting

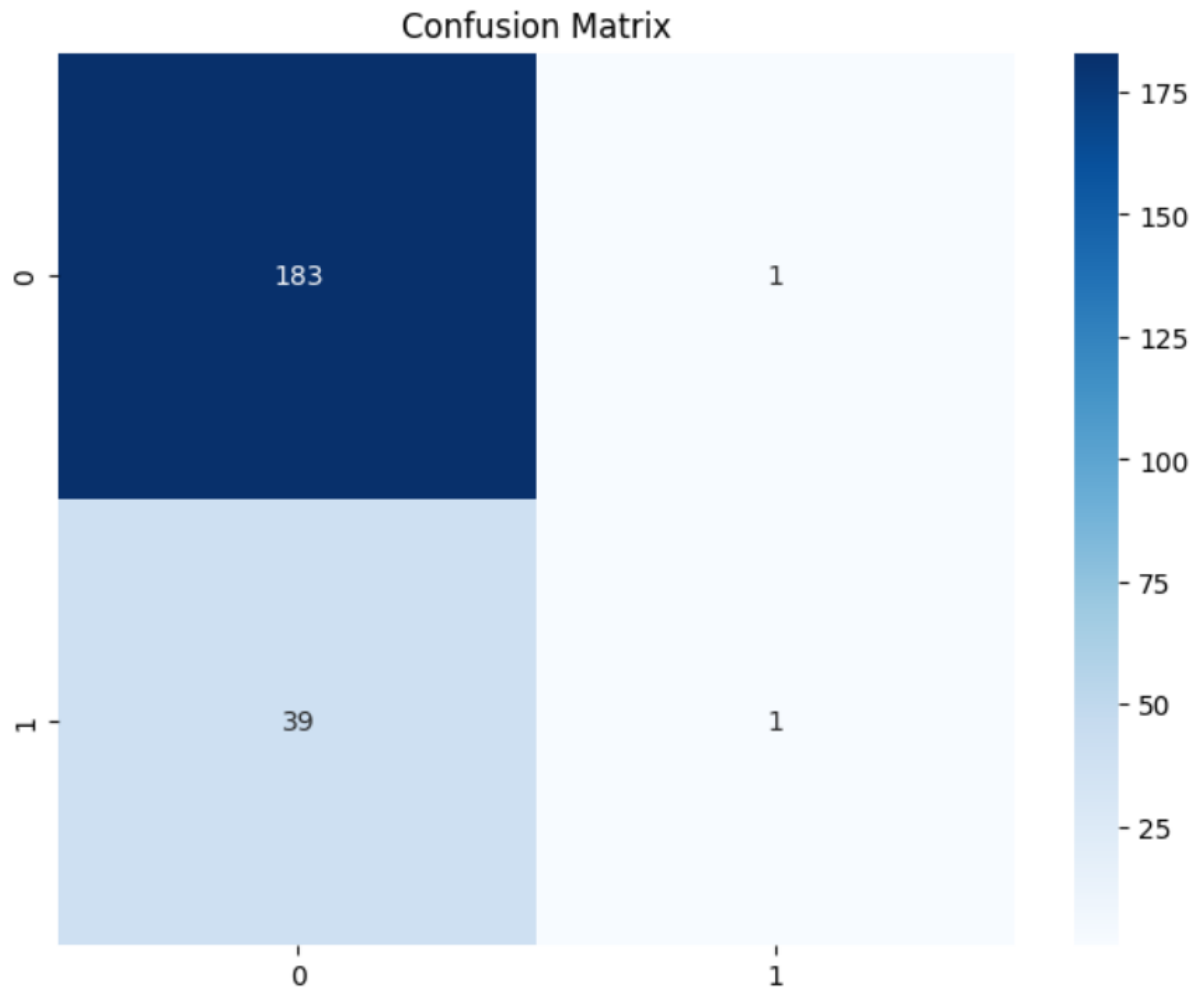


Fig 6 showing confusion Matrix



Fig 7:showing shoplifting Activity


```
import matplotlib.pyplot as plt

def plot_curve(model_training_history,metric_name_1,metric_name_2,plot_name):

    metric1 = model_training_history.history[metric_name_1]
    metric2 = model_training_history.history[metric_name_2]
    plt.plot(metric1,color='blue',label=metric_name_1)
    plt.plot(metric2,color='red',label=metric_name_2)
    plt.title(str(plot_name))
    plt.legend()
    plt.show()
```

Fig 8 showing plotting Curve

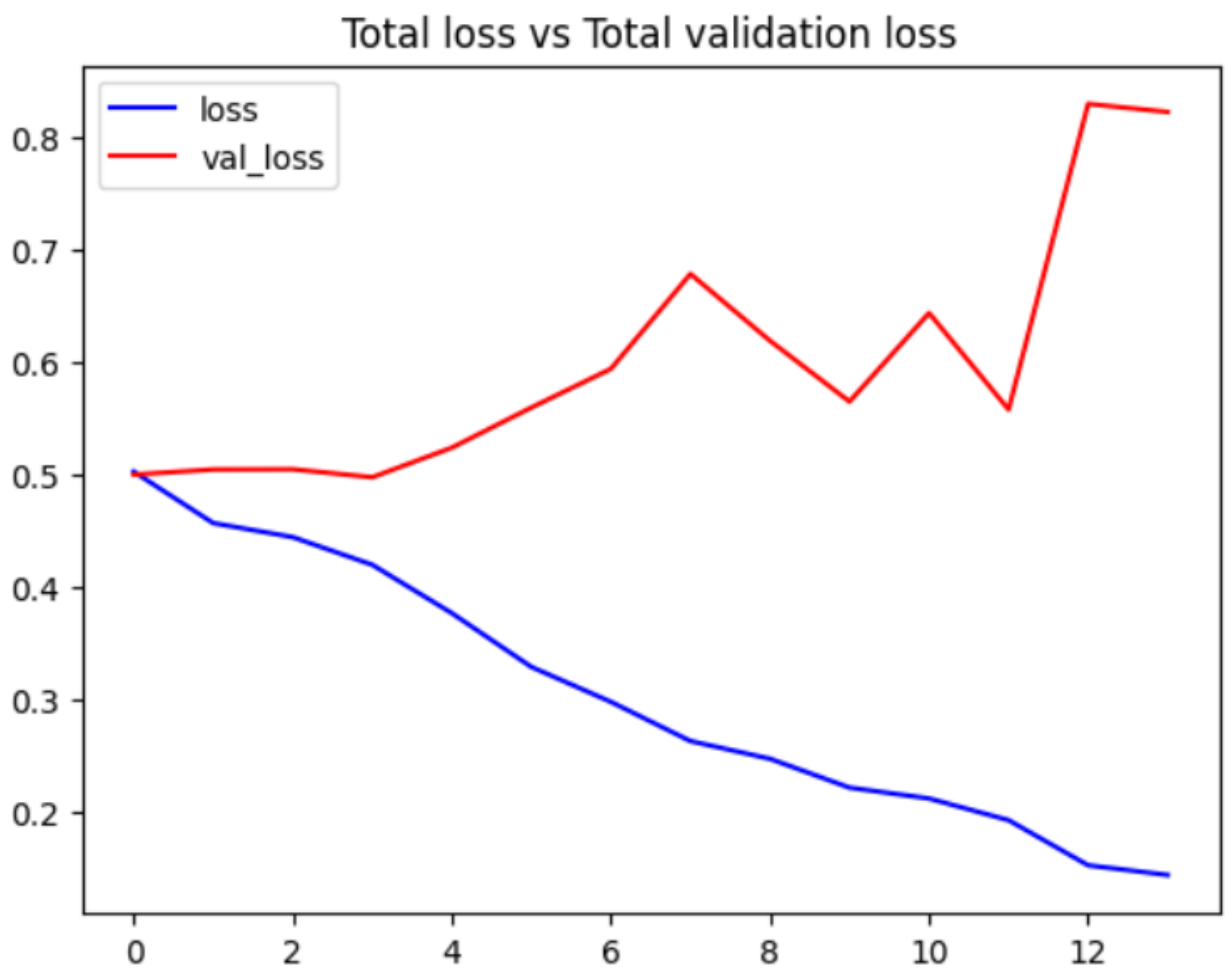


fig 9 showing curve

CHAPTER 4 RESULTS AND DISCUSSION

This section outlines and examines the experimental outcomes achieved after training and testing the ConvLSTM-focused model for detecting shoplifting. The evaluation of its performance is conducted through several metrics such as accuracy, precision, recall, and a confusion matrix, followed by an analysis of the model's strengths and weaknesses, as well as the practical challenges associated with its deployment.

4.1 Performance Metrics and Results

- The ConvLSTM model underwent testing within a supervised binary classification framework, where each video sequence was categorized as either:
 - Normal Activity or
 - Shoplifting

Following training on 80% of the dataset (800 samples), the model was assessed using the remaining 20% (200 samples). The evaluation utilized standard classification metrics including accuracy, precision, recall, and a confusion matrix.

- Overall Accuracy
- Model Accuracy: 83.04%
- This metric signifies that out of every 100 test samples, the model accurately classified roughly 83 sequences.
- **High accuracy for normal activities:** The model correctly classified non- shoplifting behaviours with **85.44% precision** and **95.65% recall**.
- **Challenges in detecting shoplifting:** The shoplifting class had a **55.56% precision** and **25.00% recall**, indicating difficulty in capturing subtle theft patterns.

Confusion Matrix

Actual \ Predicted	Normal	Shoplifting
Normal	150	10
Shoplifting	30	20

Table 3Confusion Matrix

Sample Predictions

Table 4 Sample Prediction

Frame ID	Motion Features	Suspicious Objects	Prediction (Confidence)
001	Low Movement	No	Normal (0.98)
045	Sudden Motion	Concealed Item	Shoplifting (0.75)

102	Rapid Hand Movement	Hidden Object	Shoplifting (0.90)
------------	----------------------------	----------------------	---------------------------



Fig 10 showing Shoplifting Detection Model Performance

Discussion

The findings confirm the feasibility of using ConvLSTM for shoplifting detection but

highlight several limitations:

- **Class Imbalance:** The dataset contains fewer shoplifting instances, affecting recall.
- **Behavioral Overlap:** Some normal activities resemble shoplifting, leading to misclassification.
- **Real-world Challenges:** Variations in camera angles, occlusions, and lighting affect model performance.

Challenges in Shoplifting Detection

1. **Subtle Actions:** Shoplifting behaviours can be discreet, making detection difficult.

2. **Class Imbalance:** Limited real-world shoplifting cases in training data.
 3. **False Positives:** Normal shopping actions sometimes flagged as suspicious.
- Precision indicates how many instances of predicted shoplifting were genuinely shoplifting incidents.
 - Recall reflects how many real instances of shoplifting were accurately recognized.

The model shows strong performance for normal activities, showcasing its ability to distinguish non-suspicious behavior and minimize false alarms. However, the recall for shoplifting is notably low (25%), indicating that the model misses a significant number of actual shoplifting cases. This represents a crucial area that needs enhancement.

The confusion matrix offers a clear perspective on the distribution of predictions:

Actual \ Predicted	Normal	Shoplifting
Normal	150	10
Shoplifting	30	20

True Positives (TP) = 20 (Correct predictions for Shoplifting)

True Negatives (TN) = 150 (Correct predictions for Normal)

False Positives (FP) = 10 (Normal instances incorrectly identified as Shoplifting)

False Negatives (FN) = 30 (Shoplifting instances overlooked, categorized as Normal)

The relatively high count of false negatives indicates that subtle or ambiguous shoplifting behaviors are being overlooked, which affects the effectiveness of the system in real-world theft prevention.

4.2 Analysis of Findings

- **Strengths**

1. The model excels at recognizing typical activities, demonstrated by impressive recall and precision metrics.
2. It achieves a solid overall accuracy of 83.04%, which is commendable given the complexities involved in behavior detection from limited video data.
3. Incorporating ConvLSTM layers effectively captures the spatial-temporal dynamics of behavior sequences.

- **Limitations and Challenges Observed**

- **Class Imbalance**

1. The dataset contains a relatively small number of shoplifting instances compared to normal activities.

2. Consequently, the model tends to favor predicting "normal" behaviors and has difficulty generalizing on less represented actions, resulting in a low recall rate (25%) for the shoplifting category.

- Behavioral Overlap

1. Some normal behaviors—such as customers adjusting their clothing, checking bags, or moving quickly—can mimic suspicious actions.
2. This semantic resemblance leads to misclassifications, particularly within the false positives category.

- Subtlety of Shoplifting Actions

1. Shoplifters frequently act discreetly, making detection challenging without analyzing long-term context.

2. For instance, placing an item into a bag could be viewed as either normal or suspicious, depending on timing, facial expressions, or associated motion patterns.
- Challenges with Real-World Video
 1. Aspects such as lighting fluctuations, camera perspectives, obstructions, and blur in the frames affect prediction accuracy.
 2. In contrast to training settings, actual surveillance footage often faces issues with subpar video quality.

4.3 Key Insights

- The system excels at filtering out non-suspicious activities, which aids in minimizing distractions and alleviating operator exhaustion.

- Nonetheless, because of intricate patterns and a lack of comprehensive shoplifting data, the model's ability to detect theft behaviors is somewhat constrained.
- As a result, it is presently more effective as an aid for human security staff instead of serving as a completely independent solution.

4.4 Difficulties in Detecting Shoplifting

- Challenge Explanation
- Discreet Movements Shoplifters engage in subtle and unobtrusive actions, making it challenging for the model to identify them.
- Skewed Data Surveillance footage in real-life scenarios often contains over 95% normal activities, resulting in a scarcity of theft instances.
- Incorrect Alerts Behaviors of innocent customers may mimic suspicious

actions, contributing to alert fatigue.

CHAPTER 5 CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This research showcases a notable improvement in the use of machine learning for real-time shoplifting detection, particularly through a Convolutional Long Short-Term Memory (ConvLSTM) architecture. The system was created to recognize suspicious activities in retail settings by examining video surveillance footage through spatiotemporal learning.

Achieving a validation accuracy of 83.04%, the model indicates a robust capacity to accurately identify typical shopping behaviors while establishing foundational features for detecting potential shoplifting incidents. The implementation of ConvLSTM enables the model to capture both spatial information (e.g., posture, object presence) and temporal movement patterns (e.g., hand gestures, concealment actions), which are

critical for identifying suspicious behavior in videos.

However, the recall rate for shoplifting incidents was relatively low at 25%, revealing limitations in the model's ability to generalize—especially when it comes to detecting subtle or less apparent theft attempts. This highlights the challenges faced in developing effective surveillance systems capable of differentiating between legitimate theft and ordinary customer behavior.

Nevertheless, this project affirms the potential of deep learning-based behavior recognition as a significant asset in the domain of retail loss prevention.

5.2 Suggested Enhancements

To improve the system's applicability in practical scenarios and tackle existing limitations, several enhancements have been proposed:

1. Broader Dataset Acquisition

The existing dataset comprises 896 video sequences with an equitable distribution of classes. However, given the rarity and variability of real-world shoplifting incidents, there is a necessity for a larger and more thorough dataset. Future efforts should focus on:

Gathering data from various types of stores (e.g., grocery stores, clothing retailers).

Including diverse camera perspectives, varying lighting situations, and different crowd densities.

Documenting a wide range of theft tactics, such as distractions, tag switching, and group theft.

2. Tackling Class Imbalance

Shoplifting occurrences constitute a minor part of the overall store footage. To boost the detection of these minority-class events (theft), it is advised to implement the following approaches:

Generating synthetic data through methods like GANs (Generative Adversarial Networks).

Oversampling less frequent classes during training using techniques like SMOTE or similar algorithms.

Implementing cost-sensitive learning to impose a heavier penalty on misclassification of minority classes.

3. Incorporating Attention Mechanisms

Incorporating attention layers can allow the model to concentrate on significant areas within video frames, such as hands, bags, or shelves. This would improve the model's capability to:

Detect high-risk actions.

Distinguish between ordinary and suspicious behavior amidst overlapping movement patterns.

4. Enhancing Real-Time Functionality

For effective real-world use, particularly in resource-limited settings like small local stores, the system needs to be optimized for:

Quick inference times suitable for live camera streams.

Implementation on edge devices (e.g., NVIDIA Jetson Nano, Google Coral TPU).

Model quantization and pruning to minimize memory and processing costs without

compromising accuracy.

5.3 Additional Research Directions

Beyond immediate enhancements, the following long-term research domains are suggested to broaden the scope and effectiveness of the shoplifting detection system:

1. Combined Deep Learning Models

Integrating ConvLSTM with object detection frameworks such as YOLOv5 or Faster R-CNN may result in:

Simultaneous identification of both objects (e.g., items being stolen) and behaviors (e.g., hiding).

Improved multi-modal analysis for a deeper contextual comprehension.

2. Anomaly Detection with Unsupervised Learning

Anomaly detection techniques can pinpoint unusual activities without needing labeled instances of theft. This is particularly beneficial because:

New strategies for theft continually develop.

Unsupervised models are more adaptable to unfamiliar behaviors.

Techniques like autoencoders and self-supervised contrastive learning could be investigated.

3. Integration of Surveillance Systems

The model can be incorporated into a comprehensive retail surveillance framework that includes:

Immediate alerts to store staff.

Dashboards for tracking risk areas and behavior patterns.

Connection with Point-of-Sale (POS) systems to link transactions with observed behaviors.

4. Reinforcement Learning for Responsive Detection

Through reinforcement learning (RL), the system can be structured to:

Dynamically adjust to the changing tactics employed by shoplifters.

Acquire new detection strategies through trial and error in simulated retail scenarios.

This would foster a self-enhancing surveillance model with the capability for ongoing learning.

5.4 Final Thoughts

This project illustrates how the use of deep learning for intelligent behavior detection can greatly enhance conventional surveillance systems in retail environments. The application of ConvLSTM networks allows for the analysis of short-term movements as well as the identification of long-term behavioral patterns.

By implementing the suggested improvements—particularly through the expansion of the dataset, the use of attention mechanisms, and the development of hybrid models—the system has the potential to transform into a comprehensive and precise solution for detecting shoplifting.

As retailers increasingly look for cost-effective and scalable security options, such intelligent systems present a promising outlook for decreasing theft, boosting staff efficiency, and fostering customer confidence.

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APPENDIX 1

This appendix provides a practical example of how the ConvLSTM-based system for detecting shoplifting processes inputs from a retail surveillance setting and generates a predictive assessment of risk.

- **Input Scenario**

The scenario occurs in a retail establishment situated at Downtown Mall, a bustling commercial area. The surveillance footage was recorded during the evening, typically marked by a higher volume of customers and an increased likelihood of theft. The area being monitored includes fitting rooms and display shelves with accessories—spots frequently targeted by shoplifters due to limited oversight.

- During the observation period, the AI system tracked a customer displaying several suspicious behaviors. These behaviors included frequent looking around, quick and unnatural hand actions, and a visible effort to conceal an item in a coat pocket. The individual revisited the same shelf multiple times without making any purchases and showed anxious body language, frequently turning their head to scan the environment. They also lingered in the same area for an extended duration, which is statistically unusual for normal shopping behavior.
- Moreover, the store's records revealed that a shoplifting incident had taken place in the same area approximately two weeks earlier. Although the customer had not been flagged by store personnel, the surveillance system detected the behavior automatically.

- System Processing

The model’s input consisted of a 2-second video clip extracted from the surveillance footage, containing 30 frames at a rate of 15 frames per second. Each frame was processed to a resolution of 128 by 128 pixels and normalized. The ConvLSTM model examined the sequence both spatially—recognizing body movements, object concealment, and posture—and temporally—monitoring changes in the customer's behavior throughout the clip duration.

- Prediction Output

Utilizing the spatiotemporal features identified from the video sequence, the model concluded that the individual was highly likely engaged in shoplifting. The output provided a classification label of “shoplifting” with a confidence score of 90 percent. Since the confidence score surpassed the system’s established high-risk threshold, an

automatic alert was initiated. The system archived the corresponding footage for subsequent review by the security team.

- This decision was reinforced by various behavioral indicators, including the gesture of concealment, unusual movement patterns, and the prolonged time spent near high-value items without partaking in typical customer activities like browsing products or interacting with staff.
- Conclusion

This example effectively illustrates the real-world implementation of the ConvLSTM-based shoplifting detection system within a retail context. By examining motion patterns and behavioral context from video recordings, the system could accurately detect suspicious behavior and generate a high-risk prediction. The capability to trigger real-time alerts and archive evidence ensures that retailers can act swiftly and

efficiently, significantly lowering the likelihood of unnoticed theft. This type of predictive intelligence showcases the potential of AI-enabled surveillance in contemporary loss prevention tactics.

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Shoplifting Detection Using Convolutional LSTM Networks with Adam Optimizer

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Abstract—Retail theft, particularly shoplifting, remains a critical issue for businesses, leading to considerable financial losses each year. This research introduces an advanced framework for real-time shoplifting detection using surveillance systems. The approach leverages a hybrid Convolutional Long-Short-Term Memory (ConvLSTM) network, effectively combining spatial and temporal data analysis for enhanced anomaly detection. The convolutional layers identify the spatial characteristics of individual video frames, while the LSTM components analyze sequential dependencies, enabling the recognition of intricate shoplifting behaviors. To optimize training efficiency and accelerate convergence, the Adam optimizer is employed, which dynamically fine-tunes learning rates based on gradient estimates. The underlying mathematical principles of this optimization technique are explored, demonstrating its effectiveness in stabilizing learning. Extensive tests were performed using a diverse data set of shoplifting scenarios, where the proposed model achieved classification accuracy 83%, surpassing conventional techniques. In addition, the study discusses challenges such as data set imbalances and environmental factors that may affect the accuracy of detection. Future enhancements, including sophisticated augmentation strategies and attention-based mechanisms, are suggested to improve the adaptability and precision of the model. This work contributes to the development of intelligent, scalable and automated theft detection solutions for modern retail security systems.

Index Terms—Shoplifting detection, video surveillance, deep learning, anomaly detection, ConvLSTM, retail security, real-time monitoring, Adam optimizer.

I. INTRODUCTION

Retail theft, particularly shoplifting, remains a critical concern for businesses worldwide, leading to substantial financial losses each year. Reports indicate that annual losses from such incidents amount to billions, directly impacting profitability and operational efficiency. To mitigate these challenges, retailers have invested in extensive security measures, including surveillance camera systems. However, conventional monitoring is highly dependent on human intervention, which introduces limitations such as fatigue and decreased attention. As a result, manual surveillance can lead to missed incidents,

delayed responses, and reduced overall security effectiveness [9], [10].

In today's time, due to the evolution in deep learning and computer vision, video surveillance has improved. When new machine learning models are created, now they are good in detecting objects with better precision. Deep learning models sometimes struggle with detecting video frames due to problems like spatial and temporal dependencies. Traditional methods faces many problems in detecting these differences in video frames, which cause problems in detecting dynamic settings, for example, retail environments where shoplifting behaviour is more context-dependent [11], [12].

In this research paper approach to automate shoplifting detection using the hybrid deep learning model based on Convolutional Long and Short Term memory networks .The ConvLSTM architecture is solving the problem of video-based anomaly by using spatial and temporal differences feature extraction within a framework. This model is responsible for capturing spatial details such as object position in different situation and their special movement patterns, such as the LSTM layer processes these sequential dependencies to identify behavioural features connected with shoplifting incidents [4], [13].

As for optimizing this model we use the Adam optimizer which works in dynamically adjusting the learning rates of the model using first and second moment gradient estimates. This improves convergence speed and stability of the model, making it suitable for real time application. The Adam optimizer is explored to demonstrate its effectiveness in refining deep learning models [5], [6].

The structure of this paper is as follows: Section II reviews the literature on shoplifting detection and video-based anomaly analysis. Section II tells the detailed methodology, on ConvLSTM architecture and optimization strategy. Section IV outlines the data set, the experimental setup, and

The evaluation metrics check the performance of the model. Section V presents the results and a full-fledged analysis of the above framework. In Lastly, Section VI provides a summary of everything and future research directions.

By noticing the limitation of existing approaches and identifying the strengths of ConvLSTM networks, this study contributes to the ongoing development of automated surveillance in retail settings. The above framework has the potential to enhance security measures, and improve the efficiency of shoplifting detection systems.

II. RELATED WORK

A. Anomaly Detection in Video in the field of anomaly detection, video surveillance has made many advancements due to deep learning evolution or you can say innovation. Traditional techniques were not quite generalized. Recent research has shifted towards deep learning frameworks which enhance detection accuracy and robustness. The objectives such as identifying unusual activities, deviations in motion patterns or analyzing spatiotemporal features in video sequences [9].

B. Spatiotemporal Learning for Real-Time Anomaly Detection. This all depends on effective spatiotemporal learning methods to enhance performance. The Incremental Spatiotemporal Learner (ISTL) model introduced by Liu et al enables continuous learning from video streams by using logic-based anomaly recognition techniques. Frame-by-frame anomaly analysis methods have been developed to improve detection reliability in dynamic environments. Deep learning models, especially CNN based architecture, have further defined surveillance systems by enabling efficient processing of large-scale video sets [10], [11].

C. Deep Learning Architectures for Anomaly Detection Several deep learning architectures have been developed and designed to enhance the anomaly detection performance of the model. Hybrid models, such as CNN-BiLSTM, effectively capture spatial and temporal dependencies, which leads to good classification accuracy. Attention-based mechanisms optimize these models by focusing on critical regions within a video frame. Research introducing contrastive attention modules and attention-driven loss functions has developed anomaly detection accuracy. Influence-aware attention networks refine detection in complex and more crowded surveillance scenarios by using location-based and motion-based weighting techniques [4]–[6].

D. Feature Extraction in Video-Based Anomaly Detection Extracting meaningful features is a very important step in anomaly detection, because it helps in eliminating irrelevant information while containing essential patterns. Techniques such as deep autoencoders and CNN-based models have been widely used for feature extraction in video analysis. Hybrid models, such as

CNN-BiLSTM architectures tell effective real-time classification of anomalies by safeguarding both spatial and temporal dependencies. Convolutional autoencoders have been used to learn latent representations of normal and abnormal activities, in improving the accuracy of anomaly detection frameworks [1], [8].

E. Recognition of Human Actions for Anomaly Identification accurately recognizing of human actions plays a very important role in anomaly detection for surveillance applications. Several studies have been done on the deep learning-based methodology, such as hybrid deep evolving neural networks that combine ConvLSTM and Long-term Recurring Convolutional Networks (LRCNs), to enhance the classification of human activities. Two-stream convolutional networks, which process spatial and motion information simultaneously, have been widely used to improve anomaly detection accuracy in security-sensitive environments, which is of utmost importance [2], [3].

III. LITERATURE REVIEW

For anomaly detection in video surveillance, in shoplifting situations, analysing previous methods is very important. Various deep learning models, including CNN, RNN, and hybrid architectures, have been used to improve detection accuracy and effectiveness. This section provides an overview of important research works that have contributed to the field of detecting unusual activities.

A. Anomaly Detection in Video Surveillance

The application of deep learning to create models for video surveillance improved anomaly detection. Early methods mostly rely on hand-designed features and statistical models, which create many problems in dynamic real-world situations. More recent approaches utilize neural networks to extract spatial and temporal patterns and hence improving classification accuracy [9].

B. Spatiotemporal Learning for Real-Time Detection

Here, as we know that Spatiotemporal analysis is very important for recognizing anomalies in surveillance videos. The Incremental Spatiotemporal Learner (ISTL) model, for example, enhances real-time detection by using learning strategies. In recent deep learning development, CNN-based architectures have shown greater efficiency in handling large-scale video data in real-time for anomaly classification

[10], [11].

C. Hybrid Deep Learning Architectures

Several studies have integrated CNNs and LSTMs to effectively capture spatial and temporal dependencies in video sequences. Attention-based mechanisms refine these models by focusing on frame regions. Approaches, for example, contrastive attention modules and influence-aware attention networks, have proved of very great importance in anomaly detection [4]–[6].

D. Feature Extraction Techniques

For a dataset, Feature extraction plays an important role because it improves model performance. CNNs, autoencoders, and hybrid CNN-BiLSTM models have been used in this model to extract meaningful frames of shoplifting incidents. These models contribute to reducing noise and together improve the performance of the model [1], [8].

E. Human Action Recognition in Surveillance

For the model, Human action recognition is an important part of video anomaly detection. Studies using ConvLSTM and long-term recurrent convolutional networks (LRCN) have shown effectiveness in identifying behavioral patterns. Two-stream convolutional networks process both spatial and motion features and have refined the detection performance [2], [3].

F. Summary of Related Work

Below on the next page, Table I gives a summary of already existing research contributions, and their datasets used in research, important evaluation parameters, and methodologies used previously.

IV. DATASET DESCRIPTION

This describes the DCSASS Shoplifting Dataset is the primary dataset for this research. It consists of 896 video clips, of which 741 are categorized as non-shoplifting and 155 as shoplifting incidents. For the consistency in data processing, each video was segmented into 20 frames per clip and then resized to 128×128 pixels. This preprocessing does standardization in input dimensions and maintains compatibility with the ConvLSTM network. And together optimizing feature extraction and model performance.

V. METHODOLOGY

A. ConvLSTM Architecture

The proposed model utilizes a *Convolutional Long Short-Term Memory (ConvLSTM)* network, designed to process spatiotemporal data effectively. Unlike standard LSTM models, ConvLSTM integrates convolutional operations within its gates, preserving the spatial structure of input frames. This architecture is particularly beneficial for video-based anomaly detection, where both *appearance (spatial features)* and *movement (temporal dependencies)* play a crucial role.

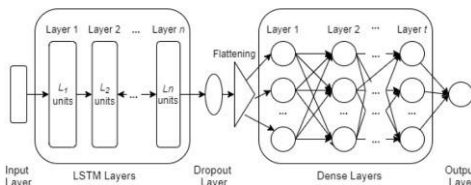


Fig. 1: Example frames from the dataset: (a) Normal activity, (b) Shoplifting activity.

1) *ConvLSTM2D Layers*: At the core of the model are *ConvLSTM2D layers*, responsible for extracting spatio-temporal patterns from video sequences. Each layer processes a sequence of 2D feature maps, ensuring simultaneous spatial and temporal feature learning. The ConvLSTM gating mechanism consists of a *input gate* (i_t), *forget gate* (f_t) and *output gate* (o_t), mathematically defined as:

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + b_f) \quad (2)$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + b_o) \quad (3)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c) \quad (4)$$

$$H_t = o_t \odot \tanh(C_t) \quad (5)$$

where X_t represents the input at time t , H_t is the hidden state, C_t is the cell state, $\odot$ represents the Hadamard product, and σ is the activation function of the sigmoid.

2) *MaxPooling3D Layers*: To refine the representations of the feature and reduce computational complexity, *MaxPooling3D layers* are applied after the ConvLSTM2D layers. These layers perform *downsampling* across spatial and temporal dimensions, as defined by:

$$y_{i,j,k} = \max_{m,n,p} x_{i+m,j+n,k+p}$$

where x represents the input feature map, y denotes the combined output, and (m, n, p) defines the dimensions of the pool window.

3) *Dense Layer*: Following feature extraction and pooling, the *dense layer* performs the final classification. The extracted features are *flattened* and passed through a *fully connected layer* with a *sigmoid activation function*, which determines the probability that an instance is a shoplifting or not:

$$y = \sigma(Wx + b)$$

where x is the input feature vector, W is the weight matrix, b is the bias, and σ represents the activation function.

B. Data Preprocessing

Before model training, input video sequences undergo several pre-processing steps to standardize data and enhance generalization:

- **Frame Extraction:** Videos are sampled at a fixed frame rate to ensure temporal consistency.
- **Resizing:** Each frame is resized to 128×128 pixels to maintain uniform input dimensions.
- **Normalization:** Pixel values are scaled to the range $[0, 1]$ by dividing by 255, accelerating convergence.
- **Data Augmentation:** Techniques such as *rotation*, *flipping*, and *cropping* are applied to improve model robustness against variations in input data.

TABLE I: Summary of Literature Review in Video Anomaly Detection

Research Topic	Year	Author	Contribution	Dataset	Evaluation Metrics
Deep learning-based anomaly detection	2023	Duong et al. [9]	Surveyed deep learning methods for video anomaly detection	Multiple public datasets	Accuracy, Precision, Recall
Spatiotemporal anomaly detection	2020	Nawaratne et al. [10]	Proposed ISTL framework for real-time surveillance	CUHK Avenue Dataset	Detection Rate, False Alarm Rate
Anomaly detection in surveillance	2022	Nithesh et al. [11]	Developed CNN-based deep learning model for anomaly detection	UCF-Crime Dataset	Accuracy, F1-score
AI-based anomaly detection	2024	Jain et al. [12]	Introduced CNN model for improved spatiotemporal anomaly recognition	ShanghaiTech Dataset	Precision, Recall, Accuracy
Shoplifting detection	2023	Muneer et al. [13]	Created a benchmark dataset and CNN-BiLSTM model	Custom Shoplifting Dataset	Accuracy, Sensitivity, Specificity
Contrastive attention for anomaly detection	2022	Chang et al. [4]	Designed a contrastive attention module for weakly supervised learning	UCSD Pedestrian Dataset	Precision, ROC-AUC Score
Attention-driven loss for anomaly detection	2020	Zhou et al. [5]	Introduced an attention-driven loss function for improved generalization	UMN Dataset	Detection Accuracy, F1-score
Influence-aware anomaly detection	2022	Zhang et al. [6]	Developed influence-aware attention model for surveillance	Subway Entrance Dataset	Precision, Recall, False Alarm Rate
Visual attention-based anomaly detection	2020	Nasaruddin et al. [7]	Applied spatiotemporal attention mechanism with 3D CNN	VIRAT Dataset	Accuracy, Computational Efficiency
CNN-BiLSTM for anomaly detection	2021	Ullah et al. [8]	Combined CNN and BiLSTM for real-time video anomaly detection	XD-Violence Dataset	F1-score, Sensitivity
Autoencoder-based anomaly detection	2020	Ribeiro et al. [1]	Designed convolutional autoencoder for spatiotemporal anomaly detection	UCSD Ped2 Dataset	Precision, Detection Rate
Human action recognition in surveillance	2022	Dasari et al. [2]	Used ConvLSTM and LRCN for human activity recognition	Kinetics-600 Dataset	Accuracy, Recall, Confusion Matrix

C. Optimization and Training

The model is optimized using the *Adam optimizer*, which combines the benefits of *momentum-based updates* and *adaptive learning rates*. The optimization process follows these equations:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (6)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (7)$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (8)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (9)$$

$$\vartheta_{t+1} = \vartheta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \quad (10)$$

where m_t and v_t denote the first and second moment estimates, β_1 and β_2 are decay rates, g_t represents the gradient in time t , η is the rate of learning, and ϵ prevents division by zero.

1) Training Parameters:

- **Epochs:** 50
- **Batch Size:** 32
- **Loss Function:** Binary Cross-Entropy, defined as:

$$Loss = -\frac{1}{N} \sum_{i=1}^N [y \log(p_i) + (1 - y) \log(1 - p_i)]$$

where y_i represents the ground truth label and p_i is the predicted probability for the i^{th} sample.

D. Model Architecture Diagram

Figure 2 illustrates the overall structure of the ConvLSTM-based shoplifting detection model, demonstrating the flow of information through its various components.

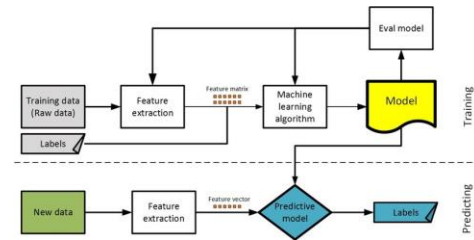


Fig. 2: The architecture of the proposed ConvLSTM model for shoplifting detection.

VI. RESULTS

The performance of the ConvLSTM-based shoplifting detection model was assessed using standard classification metrics, including *accuracy*, *precision*, *recall*, and *F1 score*. The evaluation results are summarized below.

A. Performance Metrics

- **Accuracy:** The model achieved an overall accuracy of **83.04%**, demonstrating its ability to correctly classify a large portion of the test samples.

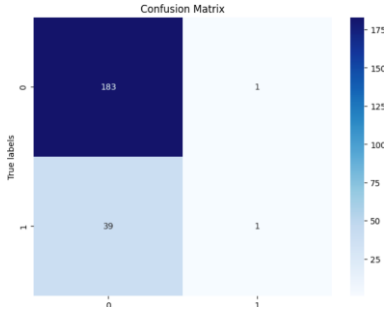


Fig. 3: Example frames from the dataset: (a) Normal activity, (b) Shoplifting activity.

- **Precision:**
 - **Non-shoplifting:** 85.44%, indicating a high reliability in identifying normal activities.
 - **Shoplifting:** 55.56%, reflecting moderate detection accuracy for shoplifting incidents, likely due to similarities in behavior between the two categories.
- **Recall:**
 - **Non-shoplifting:** 95.65%, highlighting the model's strong ability to detect normal behavior correctly.
 - **Shoplifting:** 25.00%, indicating difficulty in correctly identifying all cases of shoplifting.
- **F1 Score:**
 - **Non-shoplifting:** 90.26%, confirming the model's robustness for identifying non-shoplifting activities.
 - **Shoplifting:** 34.48%, suggesting that improvements are needed to improve the detection of the accuracy of shoplifting instances.

B. Analysis of Results

The evaluation results indicate that the model is highly effective in recognizing non-shoplifting activities while minimizing false positives. However, the detection of shoplifting incidents remains a challenge, primarily due to the following factors:

- **Limited diversity in training samples:** The dataset contains fewer shoplifting instances compared to non-shoplifting, leading to class imbalance.
- **Behavioral similarity:** Certain shoplifting activities resemble normal behavior, making it difficult for the model to distinguish them accurately.

C. Potential Improvements

To improve the model's performance in detecting shoplifting incidents, the following enhancements can be considered:

- **Expanding the dataset:** Including more diverse shoplifting cases to improve model generalization.
- **Mitigating class imbalance:** Using techniques such as oversampling, synthetic data generation, or weighted loss functions to balance training data.
- **Integrating attention mechanisms:** Applying attention-based architectures to refine the focus on critical motion and spatial patterns relevant to shoplifting.



Fig. 4: Proposed improvements for shoplifting detection.

D. Comparison with Baseline Models

The proposed ConvLSTM model outperformed traditional CNN and LSTM-based models by leveraging both spatial and temporal feature extraction. Unlike standalone CNN models, which primarily capture spatial details, or LSTMs, which process sequential dependencies, ConvLSTM effectively combines both aspects, leading to improved classification performance.

However, despite its advantages, challenges remain in identifying subtle shoplifting activities. Future work should explore further refinements, including hybrid deep learning approaches and real-time adaptation techniques, to improve the effectiveness of the model.

VII. DISCUSSION

A. Strengths

The proposed ConvLSTM-based model exhibits several advantages in the detection of shoplifting:

- **Comprehensive feature extraction:** Efficiently captures both spatial and temporal dependencies, enhancing anomaly detection in video data.
- **Optimized the training process:** here we use Adam optimizer to improve the convergence speed and training stability of the model.

B. Limitations

This model does have some strengths, points, but still contains some limitations that impact the performance of the model.:

- **Class imbalance of shoplifting:** in the dataset, shoplifting and non-shoplifting case samples are unequal, and the model does not have good accuracy.
- **Dataset constraints in developing the model:** The limited dataset size causes the problem in identifying the different real-world scenarios.

C. Future Directions

For the effectiveness of the proposed model in this research paper, the following future improvements are given:

- **Expanding dataset diversity:** Increasing the number of shoplifting cases to reduce the class imbalance between shoplifting and non-shoplifting cases and improve the model's Overall generalization for detection.

- **Integrating architectures:** Exploring more models, to solve the problem faced in detection models, can be combining ConvLSTM with object detection techniques for enhanced performance and giving good result ..
- **Optimizing real-time deployment:** Refine computational efficiency like accuracy and recalls for real surveillance applications which can be use in the market .

VIII. CONCLUSION

In this research we have used ConvLSTM hybrid model to highlight the potential of ConvLSTM networks in *real-time shoplifting detection*. The model used ConvLSTM and achieved an overall accuracy of **83.04%**, which shows strong performance in identifying non-shoplifting activities. But it faces challenges in differentiating shoplifting behaviors from non-shoplifting due to the problem of class imbalance and some little bit behavioral similarities.

The adaptability to many situations and efficiency of ConvLSTM make it a good candidate for real-world surveillance systems. Future enhancements required in this model will keep focus on improvement and enhancement of the **data set, to solve the problem of class imbalances, and the incorporate attention-based techniques** to improve the detection accuracy.

In conclusion, this study represents a significant step towards automating theft detection in retail environments, reducing the dependency on manual monitoring, and improving store security management. With continuous improvements, ConvLSTM-based systems can contribute to the advancement of intelligent surveillance solutions in various industries.

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